

Bucket Versioning, Replication and Static website Hosting in S3

Class No:6

1.Bucket Versioning in AWS S3

- **Definition:** A feature in Amazon S3 that allows you to keep multiple versions of an object (file) in the same bucket.
- **Purpose:** Protects against accidental overwrites and deletions by maintaining history.
- **Behaviour:** When versioning is enabled, every time you upload or modify an object, S3 creates a new version with a unique version ID.

Purpose of Versioning

- **Data Protection:** Prevents permanent loss if files are accidentally deleted or overwritten.
- **Recovery:** You can restore older versions of objects.
- **Audit & Tracking:** Maintains history of changes for compliance or debugging.
- **Safe Collaboration:** Multiple users can work on the same bucket without losing data.

Delete Marker

- **Definition:** A special placeholder object created when you delete an object in a versioned bucket.
- **Purpose:** It doesn't remove the object permanently; instead, it makes the latest version appear as "deleted."
- **Behaviour:**
 - When you delete an object, S3 adds a **delete marker** as the newest version.
 - The object is hidden from normal view but older versions are still accessible.
 - Removing the delete marker restores visibility of the object.

Hands on Bucket Versioning:

Step 1: While creating a S3 bucket, Turn on versioning for your S3 bucket.

Bucket name [Info](#)

bucketversioningpractice019876

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership
Bucket owner enforced

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☒ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☒ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☒ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☒ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☐ Disable

☒ Enable

Step 2: Upload Object in the created bucket with same name for two times. Each upload creates a new version with a unique ID. Updates don't overwrite; they create new versions.

(**Note:** The versions can be seen only the show versions is enabled.)

bucketversioningpractice019876 [Info](#)

Objects

Metadata

Properties

Permissions

Metrics

Management

Access Points

Objects (2)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Find objects by prefix

Show versions

☐	Name	Type	Version ID	Last modified	Size	Storage class
☐	 amber.jpeg	jpeg	ZdDdbgz0oNzy2GCqPq7WoVfXx_i1N6Fs	March 16, 2026, 18:51:52 (UTC+05:30)	17.0 KB	Standard
☐	 amber.jpeg	jpeg	T7ET7ubUjipizKUMbOejP4D6pXoXkiS	March 16, 2026, 18:51:31 (UTC+05:30)	17.0 KB	Standard

Step 3: Delete Object. Instead of removing data, S3 adds a **delete marker**.

The show versions must be disabled while deleting the objects , then enabled after deletion to see the delete marker.

bucketversioningpractice019876 [Info](#)

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Objects (3)

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Create folder

Upload

Find objects by prefix

Show versions

☐	Name	Type	Version ID	Last modified	Size	Storage class
☐	 amber.jpeg	Delete marker	UydG9B0T4xyH4dHhz9DxdAOEzEFyjDV7	March 16, 2026, 18:57:25 (UTC+05:30)	0 B	-
☐	 amber.jpeg	jpeg	ZdDdbgz0oNzy2GCqPq7WoVfXx_i1N6Fs	March 16, 2026, 18:51:52 (UTC+05:30)	17.0 KB	Standard
☐	 amber.jpeg	jpeg	T7ET7ubUjipizKUMbOejP4D6pXoXkiS	March 16, 2026, 18:51:31 (UTC+05:30)	17.0 KB	Standard

Step 4: Restore Object by removing the delete marker → the previous version becomes the latest and visible again.

bucketversioningpractice019876 [Info](#)

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2. Bucket Replication in AWS S3

- **Definition:** Replication is the process of automatically copying objects from one S3 bucket (source) to another (destination).
- **Purpose:** Ensures redundancy, disaster recovery, compliance, and low-latency access by keeping copies of data in different locations.
- **Requirement:** Both source and destination buckets must have **versioning enabled**.

Types of Replication

1. Cross-Region Replication (CRR)

- Copies objects across different AWS regions.
- Useful for disaster recovery, compliance with data residency laws, and serving global users.

2. Same-Region Replication (SRR)

- Copies objects within the same AWS region.
- Useful for maintaining copies in different accounts, compliance, or separating workloads.

3. Cross-Account Replication (CAR)

- Copies objects for buckets in two different accounts.

Replication Rules

Replication rules define **what to replicate** and **how**. Key points:

- **Scope:** You can replicate the entire bucket or specific prefixes/tags.
- **Ownership:** Replication can be set up between buckets in the same or different AWS accounts.
- **IAM Role:** A replication IAM role must be created to grant S3 permission to replicate objects.
- **Options:**
 - Replicate existing objects or only new ones. (we used this in our hands-on)
 - Replicate delete markers (optional).
 - Replicate specific metadata (like ACLs, tags).
- **Versioning:** Required on both source and destination buckets.

Hands on Bucket replication:

Step 1: Create two Buckets. Source bucket: **source-bucket-098769** and Destination bucket: **destinatio-bucket-098769**. They can be public or private. But, version must be enabled for both.

To upload files and folders, or to configure additional bucket settings, choose **View details**.

Buckets

General purpose buckets **All AWS Regions** Directory buckets

General purpose buckets (2) [Info](#)

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	Creation date
destinatio-bucket-098769	Europe (Stockholm) eu-north-1	March 16, 2026, 19:12:03 (UTC+05:30)
source-bucket-098769	Europe (Stockholm) eu-north-1	March 16, 2026, 19:11:23 (UTC+05:30)

Account Updated Storage L

Extern Updated External a public acc

Step 2: Now open the source bucket add some files, click **Management**. Follow the configurations listed below:

- (i) Click the Create Replication rule.

source-bucket-098769

ObjectsMetadataPropertiesPermissionsMetricsManagementAccess Points

Lifecycle configuration
To manage your objects so that they are stored cost effectively throughout their lifecycle, configure their lifecycle. A lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. Lifecycle rules run once day.

Lifecycle rules
Use lifecycle rules to define actions you want Amazon S3 to take during an object's lifetime such as transitioning objects to another storage class, archiving them, or deleting them after a specified period of time. [Learn more](#)

View detailsEditDeleteActionsCreate lifecycle rule

Lifecycle rule name	Status	Scope	Current version actions	Noncurrent versions actions	Expired object delete mar...	Incomplete multipart upl.
No lifecycle rules There are no lifecycle rules for this bucket. Create lifecycle rule						

Replication rules (0)
Use replication rules to define options you want Amazon S3 to apply during replication such as server-side encryption, replica ownership, transitioning replicas to another storage class, and more. [Learn more](#)

View detailsEdit ruleDeleteActionsCreate replication rule

Replication rule name	Status	Destination bucket	Destination Region	Priority	Scope	Storage class	Replica owner	Replication Time Control	KMS-encrypted objects (SSE-KMS or DSSE-KMS)	Replica modification sync
No replication rules You don't have any rules in the replication configuration. Create replication rule										

- (ii) Provide a rule name: **newruletotransferobjects** , then enable the status , choose the bucket in this same account. Then search and enter the destination account , choose create new role. Save.

Create replication rule [Info](#)

Replication rule configuration

Replication rule name

newruletotransferobjects

Up to 255 characters. In order to be able to use CloudWatch metrics to monitor the progress of your replication rule, the replication rule name must only contain English characters.

Status

Choose whether the rule will be enabled or disabled when created.

- ☒ Enabled
☐ Disabled

Priority

The priority value resolves conflicts that occur when an object is eligible for replication under multiple rules to the same destination. The rule is added to the configuration at the highest priority and the priority can be 0

Source bucket

Source bucket name

source-bucket-098769

Source Region

Europe (Stockholm) eu-north-1

Choose a rule scope

- ☐ Limit the scope of this rule using one or more filters
☒ Apply to all objects in the bucket

Destination

Destination

You can replicate objects across buckets in different AWS Regions (Cross-Region Replication) or you can replicate objects across buckets in the same AWS Region (Same-Region Replication). You can also specify a different Amazon S3 pricing tier.

- ☒ Choose a bucket in this account
☐ Specify a bucket in another account

Bucket name

Choose the bucket that will receive replicated objects.

destinatio-bucket-098769

Destination Region

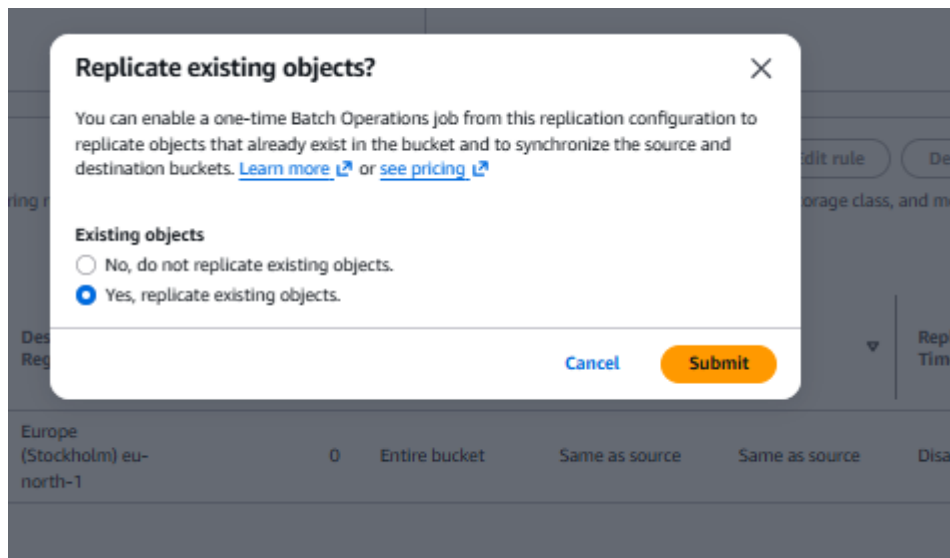
Europe (Stockholm) eu-north-1

IAM role

Permission to access the specified resources

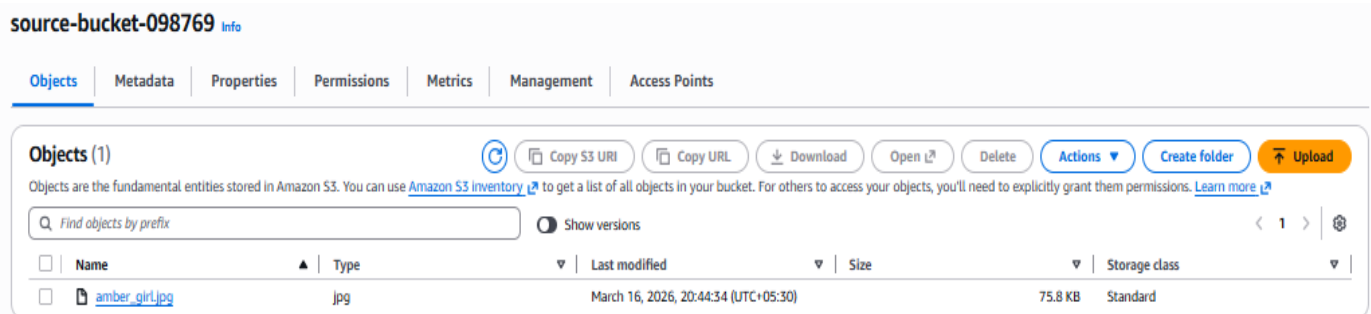
- ☒ Create new role
☐ Choose from existing IAM roles
☐ Enter IAM role ARN

(iii) It will ask for permission. Mark it as yes and submit.

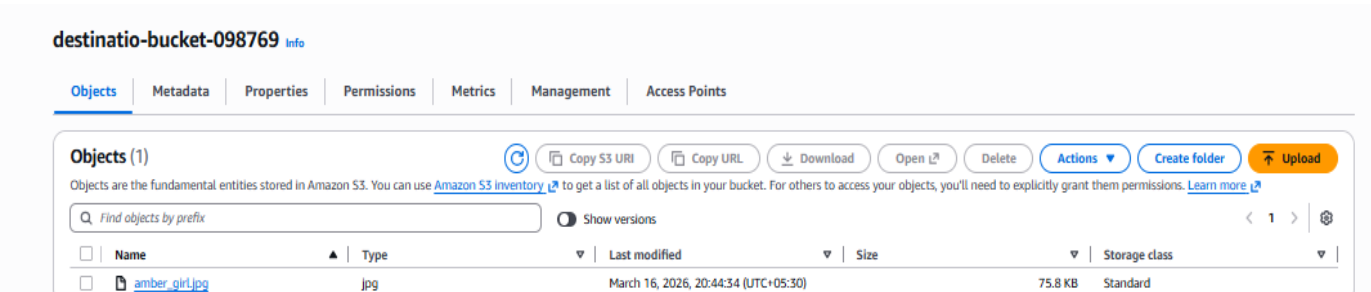


Step 3: Add objects in source bucket and wait for few minutes we can see the object is added automatically in the Destination bucket.

Source bucket:



Destination bucket:



3.Static Web Hosting in S3

- **Definition:** Amazon S3 can serve static content (HTML, CSS, JavaScript, images) directly from a bucket as a website.
- **Purpose:** Host lightweight websites without needing servers, ideal for portfolios, documentation, or landing pages. This is known as serverless. (AWS handles scaling and management)
- **Limitation:** Only supports **static content** — no server-side scripting (like PHP, Node.js).

Purpose of Static Hosting

- **Cost-effective:** Pay only for storage and data transfer.
- **Scalable:** Automatically handles large traffic without extra configuration.
- **Simple setup:** No need to manage servers or infrastructure.

Mandatory Things for Static Hosting

- **Bucket must be public** (or objects must be publicly readable) for the website to be accessible.
- **Index document** must be defined (e.g., index.html).

Hands on Steps to Enable Static Hosting

Step 1: Create an S3 Bucket. Name: staticwebhostingbucket768598454967 , make it public by enabling ACL and disable the public access deny option.

Bucket namespace

Choose the namespace where you want to create your bucket. [Learn more](#)

☒ Global namespace

By default, S3 creates general purpose buckets in the global namespace.

☐ Account Regional namespace (recommended)

General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name

Info

staticwebhostingbucket768598454967

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

Object Ownership

Info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☐ ACLs disabled (recommended)

All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☒ ACLs enabled

Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

⚠ We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. Using a bucket policy instead of ACLs to share data with users simplifies permissions management and auditing.

Object Ownership

☒ Bucket owner preferred

If new objects written to this bucket specify the bucket-owner-full-control canned ACL, they are owned by the bucket owner. Otherwise, they are owned by the object writer.

☐ Object writer

The object writer remains the object owner.

ⓘ If you want to enforce object ownership for new objects only, your bucket policy must specify that the bucket-owner-full-control canned ACL is required for object uploads. [Learn more](#)

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ Block all public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ Block public access to buckets and objects granted through new access control lists (ACLs)

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to buckets and objects granted through any access control lists (ACLs).

☐ Block public access to buckets and objects granted through any access control lists (ACLs)

S3 will ignore all ACLs that grant public access to buckets and objects.

☐ Block public access to buckets and objects granted through new public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ Block public and cross-account access to buckets and objects through any public bucket or access point policies

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

⚠ Turning off block all public access might result in this bucket and the objects within becoming public

AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☒ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

Step 2: Upload Website Files. Upload HTML, CSS, JS, and image files into the bucket. Make sure the path is same, don't change it. The name of the html file should be index.html.

Step 3: Enable Static Website Hosting

- Go to **Properties** → **Static website hosting** → **Enable**.

Edit static website hosting [Info](#)

Static website hosting
Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting
☐ Disable
☒ Enable

Hosting type
☒ Host a static website
Use the bucket endpoint as the web address. [Learn more](#)
☐ Redirect requests for an object
Redirect requests to another bucket or domain. [Learn more](#)

For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document
Specify the home or default page of the website.

Error document - optional
This is returned when an error occurs.

Redirection rules - optional
Redirection rules, written in JSON, automatically redirect webpage requests for specific content. [Learn more](#)

1

JSON Ln 1, Col 1 Errors: 0 Warnings: 0

Cancel Save changes

Step 4: Set Bucket Policy for Public Access. By default, S3 bucket objects are private. Click all -> go to actions, make all public using ACL enabled.

staticwebhostingbucket768598454967 [Info](#)

Objects (13/13) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Find objects by prefix

☒	Name	Type	Last modified	Size
☒	app.js	js	March 16, 2026, 21:36:47 (UTC+05:30)	
☒	aws-cert.jpg	jpg	March 16, 2026, 21:36:48 (UTC+05:30)	
☒	deloitte-cert.jpg	jpg	March 16, 2026, 21:36:49 (UTC+05:30)	
☒	index.html	html	March 16, 2026, 21:36:50 (UTC+05:30)	
☒	jpmorgan-cert.jpg	jpg	March 16, 2026, 21:36:50 (UTC+05:30)	
☒	Mypicture.jpeg	jpeg	March 16, 2026, 21:36:51 (UTC+05:30)	1
☒	NITHYA_SRI_PALANIVEL_Software Developer.pdf	pdf	March 16, 2026, 21:37:04 (UTC+05:30)	
☒	package-lock.json	json	March 16, 2026, 21:37:05 (UTC+05:30)	
☒	package.json	json	March 16, 2026, 21:37:04 (UTC+05:30)	315.0 B Standard
☒	pega-cert.jpg	jpg	March 16, 2026, 21:37:06 (UTC+05:30)	64.9 KB Standard
☒	server.js	js	March 16, 2026, 21:37:06 (UTC+05:30)	670.0 B Standard
☒	style.css	css	March 16, 2026, 21:37:07 (UTC+05:30)	1.9 KB Standard
☒	vista-ai-cert.jpg	jpg	March 16, 2026, 21:37:07 (UTC+05:30)	75.3 KB Standard

Download as

Share with a presigned URL

Calculate total size

Copy

Move

Initiate restore

Query with S3 Select

Edit actions

Rename object

Edit storage class

Edit server-side encryption

Edit tags

Make public using ACL

Step 5: Access Website Endpoint. AWS provides a endpoint under the properties tab of the bucket. (scrool down, it will be available in the static web hosting.)

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

We recommend using AWS Amplify Hosting for static website hosting
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. [Learn more about Amplify Hosting](#) or [View your existing Amplify](#)

S3 static website hosting

Enabled

Hosting type

Bucket hosting

Bucket website endpoint

When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)

<http://staticwebhostingbucket768598454967.s3-website.eu-north-1.amazonaws.com>

Step 6: Copy and paste the url in a browser, we can see the contents of the uploaded file.(Here I used my portfolio files.)

Not secure staticwebhostingbucket768598454967.s3-website.eu-north-1.amazonaws.com

AMETRIC... EVS CHAP 1.pdf Dell View Ads | VuexyB... Gmail YouTube Maps New Tab file systems in os -... Input Buffering in... KRCE ALUM



NITHYA SRI PALANIVEL

She/Her

EX - TCS (British Airways Project) | Software Developer | Passionate about Development & AWS |



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